CW-200 COOLING-WATER SYSTEM - DATA SHEET (sheet 1 of 2)

Arrangement. Pumps P-201A, P-201B, P-201C and P-201D take suction from the open sump (water level EL 0.0 m) and discharge in parallel, each through its own check valve, into the manifold whose outlet feeds CW-201 (isometric sheet 1). CW-201 joins the DN200 ring main CW-202 at R1. Branches leave the ring at tees Ta (CW-203: FV-201, E-201, drain), Tb (CW-204: riser to break tank TK-1), Tc (CW-205: E-202, drain), Td (CW-206: FV-203, E-203, drain) and Te (CW-207: HV-204, E-204, drain). TK-1 is open to atmosphere; the riser enters it horizontally through the tank wall, below the water surface; its overflow weir returns to the sump. From the TK-1 bottom outlet, CW-208 runs to node N5, which feeds CW-209 (E-205, drain), CW-210 (HV-206, E-206, drain) and the outfall CW-211 to the river pond. Drains and the pond outfall discharge to atmosphere at the elevation of the line end shown on the isometrics. The riser discharges into TK-1 below the water surface: its exit loss is taken against the TK-1 water level. PI-100 is on the ring at R1.

Pumps: head H (m) = H0 - k Q^2, Q in m3/s. P-201A: H0 52.0, k 1900. P-201B: H0 50.0, k 1750. P-201C: H0 47.0, k 2300. P-201D: H0 50.0, k 1800. Each pump's check valve and discharge fittings: K 4.0 on the velocity of that pump's flow in DN150. Suction losses are negligible; the manifold itself is lossless.

Pipe inside diameters: DN200 202.7 mm; DN150 154.1 mm; DN100 102.3 mm; DN80 77.9 mm. Roughness 0.045 mm. Friction: Darcy-Weisbach with f = 0.25 / [log10(e/(3.7 D) + 5.74 / Re^0.9)]^2.

Minor losses, each K applied to the velocity head of the line in which it sits:
- elbow (each change of direction along a line, counted from the isometrics) 0.3; a change of direction at R1, at a tee or at N5 belongs to that junction, not to an elbow;
- gate valve (GV) or hand valve (HV) fully open 0.15;
- flow leaving the ring or N5 into a branch line: 1.0 on the branch velocity; no loss for flow continuing along the ring, for flow from CW-201 into the ring at R1, or for flow from CW-208 into N5;
- TK-1 outlet entrance 0.5 (CW-208);
- discharge to atmosphere at a drain or at the pond, or into TK-1 below its water surface: 1.0;
- exchangers: E-201 60, E-202 26, E-203 34, E-204 22, E-205 60, E-206 55;
- control valves FV-201 and FV-203, K versus opening: 20 % 420; 30 % 170; 35 % 115; 45 % 55; 60 % 22; 80 % 9.0; 100 % 4.5.

Controllers: FIC-201 (on FV-201) and FIC-203 (on FV-203). In AUTO a controller holds its line's flow at the setpoint if its valve can do so; if even fully open its valve cannot reach the setpoint, it sits fully open. In MANUAL the valve stays at the opening the operator set.

TK-1: weir crest EL 35.0 m, rectangular, length 0.8 m, overflow Q = 1.84 L h^1.5 (h = level above crest, m). Bottom outlet nozzle EL 25.0 m.

Water: density 998.2 kg/m3, kinematic viscosity 1.004e-6 m2/s, vapour pressure 2.339 kPa abs. Atmospheric pressure 101.325 kPa. g 9.81 m/s2.

Notes.
1. Report steady-state values for the line-up in force at the end of the log.
2. Water cannot sustain an absolute pressure below its vapour pressure. Where a full-pipe solution would require one, the column separates at the point of lowest pressure: the pressure there equals the vapour pressure, the line downstream runs partly full under vapour, and its drop adds no suction.
3. For the pressure at a point, include the losses of every fitting upstream of the point and count a fitting located at the point as upstream.
4. PI-100 reading = energy head at R1 minus the R1 elevation minus the velocity head in ring segment R1-Ta, in bar gauge.
5. Pump and valve states change only through operations recorded in the log. The PI-100 low alarm is set at 3.0 bar g; a pump in AUTO-standby starts automatically if PI-100 falls below 3.0 bar g. Only an operator stops a running pump.
6. At every junction (R1, each tee, N5) the energy head (elevation + pressure head + velocity head) is the same in every connected line, apart from the losses listed above. Report steady-state values to the precision asked; the rules above define them exactly.

CW-200 OPERATING LOG - NIGHT SHIFT 14/15 OCT (sheet 2 of 2)

22:00 Handover. Running P-201A, P-201B. P-201C AUTO-standby. P-201D out of service (motor bearing replacement, permit 8102). FIC-201 AUTO SP 18.0 L/s; FIC-203 AUTO SP 16.0 L/s. HV-204, HV-206 open.

22:40 Rounds. All exchanger outlet temperatures normal.

23:20 P-201B tripped (motor overload). PI-100 low alarm; P-201C started on standby logic.

23:35 Called electrician for P-201B. PI-100 low alarm still standing; no standby pump available.

00:15 Sump makeup checked, level normal.

01:10 FT-203 signal failed (bad quality). FIC-203 to MANUAL, FV-203 output set to 35 %.

01:40 Permit 8117: E-204 tube-side clean. HV-204 closed and locked. PI-100 low alarm cleared.

02:30 Electrician reset P-201B overload; megger test good. P-201B selected AUTO-standby.

03:15 Rounds. TK-1 level gauge glass and weir box inspected.

03:50 Permit 8102 closed: P-201D bearing job complete. P-201D selected AUTO-standby.

04:30 P-201C high vibration alarm 7.1 mm/s. Started P-201B from control room, then stopped P-201C. P-201C selected OFF for vibration analysis.

05:20 E-206 suspected tube leak (conductivity high on outlet). HV-206 closed to isolate E-206. Maintenance informed.

06:00 Permit 8117 closed: E-204 cleaning complete. HV-204 unlocked and opened; E-204 back in service.

06:45 Instrument tech replaced FT-203. FIC-203 left in MANUAL at 35 % pending loop check by day shift.

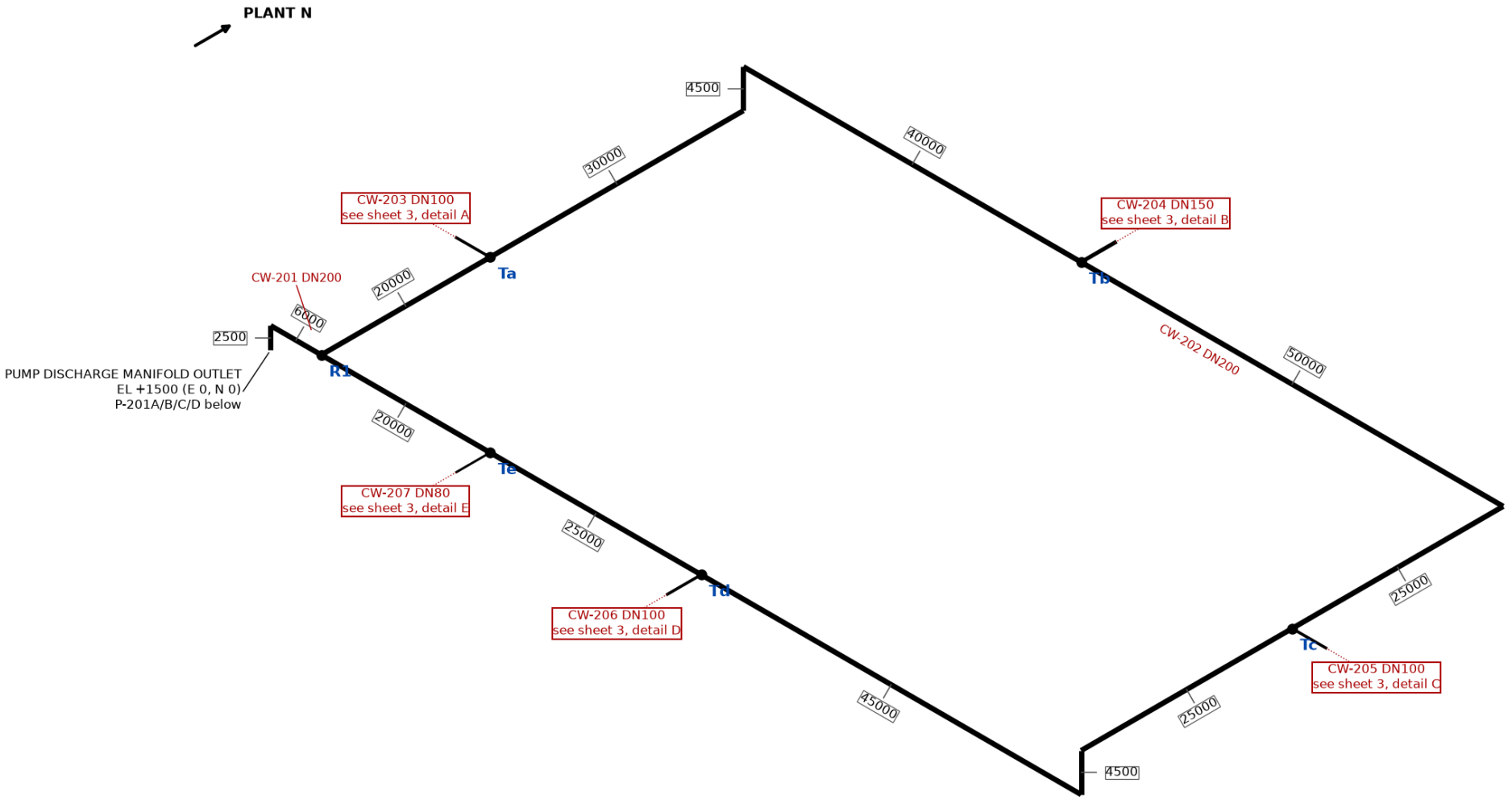
07:10 Process engineer request: FIC-201 setpoint raised to 26.0 L/s (FIC-201 remains AUTO).

07:40 P-201D 5-minute test run from local station for bearing run-in, then stopped locally. Returned to AUTO-standby.

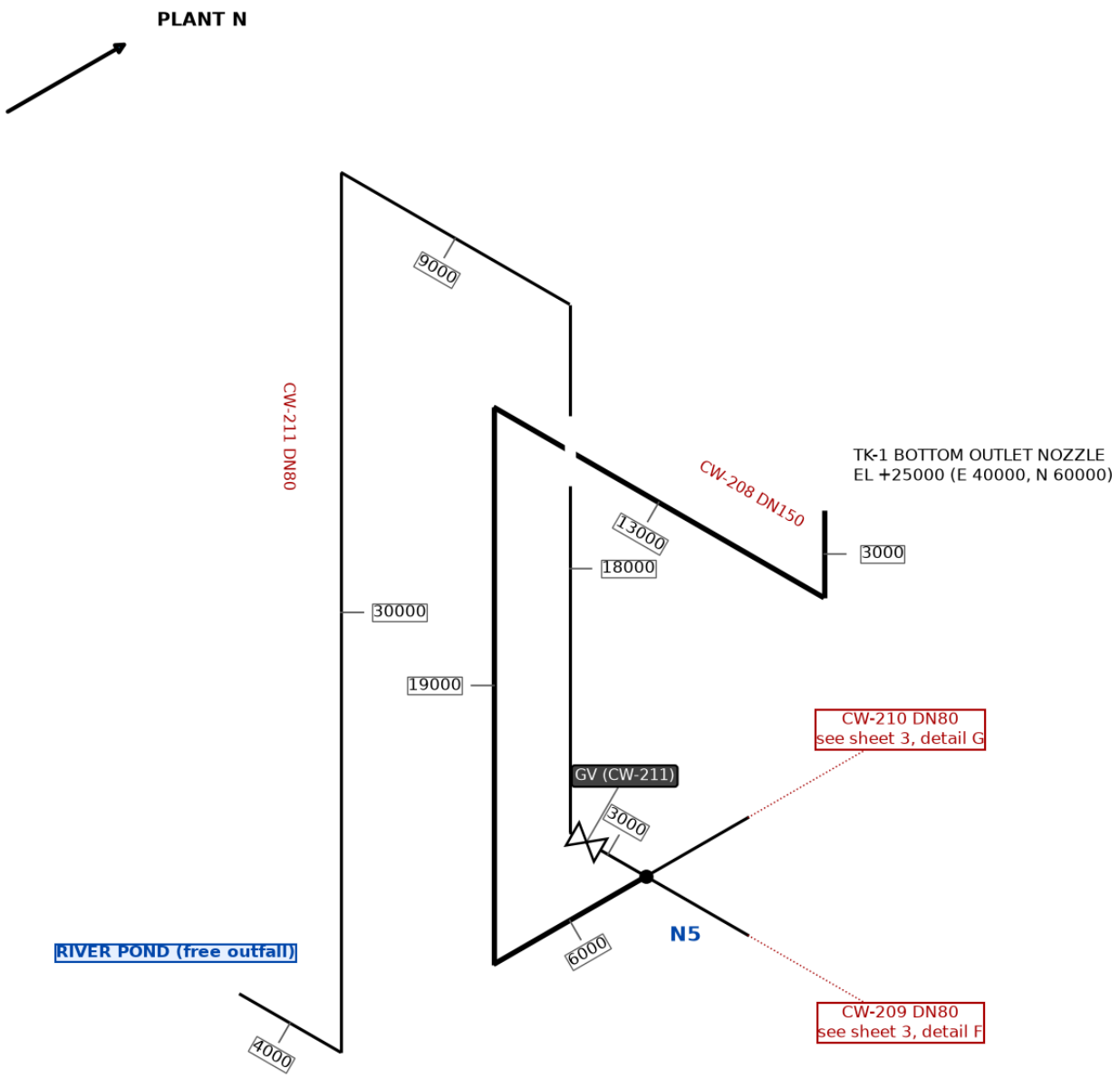
08:20 Rounds. No further changes.

09:00 Handover to day shift.

CW-200 isometric, sheet 1 of 3: pump manifold, DN200 ring main and its branches. To scale; dimensions in mm along centrelines. Plant north up-right.

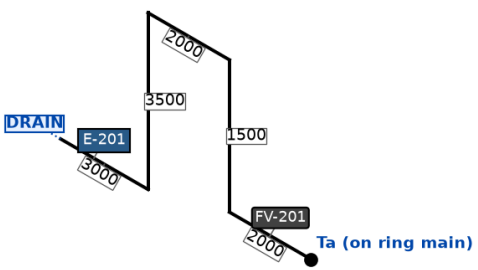


CW-200 isometric, sheet 2 of 3: TK-1 gravity side. To scale; dimensions in mm along centrelines. Plant north up-right.

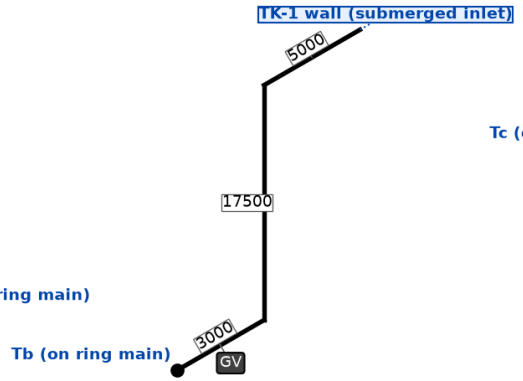


CW-200 isometric, sheet 3 of 3: branch details. Dimensions in mm along centrelines.

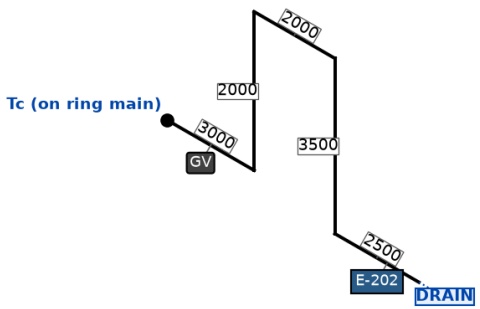
Detail A: CW-203 DN100 (not to scale)



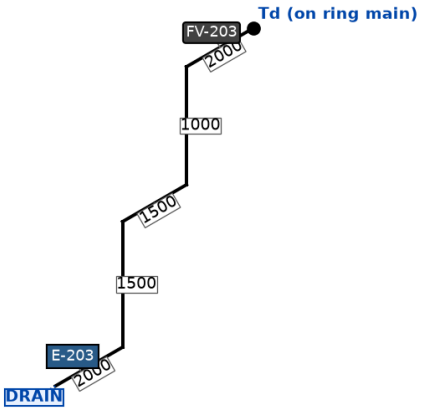
Detail B: CW-204 DN150 (not to scale)



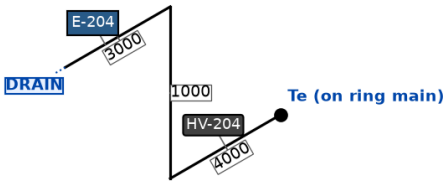
Detail C: CW-205 DN100 (not to scale)



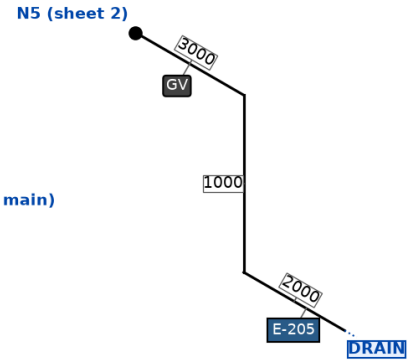
Detail D: CW-206 DN100 (not to scale)



Detail E: CW-207 DN80 (not to scale)



Detail F: CW-209 DN80 (not to scale)



Detail G: CW-210 DN80 (not to scale)

